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Aim: To perform Hierarchical Clustering using agglomerative methods and interpret dendrograms for cluster formation.

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
path="/content/drive/MyDrive/ML datasets/Spotify_Songs_Dataset.csv"
df = pd.read_csv(path)
df.head()
```

	Unnamed: 0	track_id	artists	album_name	track_name	popularity
0	0	5SuOikwiRyPMVoIQDJUgSV	Gen Hoshino	Comedy	Comedy	73
1	1	4qPNDBW1i3p13qLCt0Ki3A	Ben Woodward	Ghost (Acoustic)	Ghost - Acoustic	55
2	2	1iJBSr7s7jYXzM8EGcbK5b	Ingrid Michaelson;ZAYN	To Begin Again	To Begin Again	57
3	3	6lfxq3CG4xtTiEg7opyCyx	Kina Grannis	Crazy Rich Asians (Original Motion Picture Sou...	Can't Help Falling In Love	71
4	4	5vjLSffimiIP26QG5WcN2K	Chord Overstreet	Hold On	Hold On	82

5 rows × 21 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 114000 entries, 0 to 113999
Data columns (total 21 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
```

```
 0   Unnamed: 0        114000 non-null  int64
 1   track_id          114000 non-null  object
 2   artists           113999 non-null  object
 3   album_name        113999 non-null  object
 4   track_name        113999 non-null  object
 5   popularity        114000 non-null  int64
 6   duration_ms       114000 non-null  int64
 7   explicit          114000 non-null  bool
 8   danceability      114000 non-null  float64
 9   energy            114000 non-null  float64
 10  key               114000 non-null  int64
 11  loudness          114000 non-null  float64
 12  mode              114000 non-null  int64
 13  speechiness       114000 non-null  float64
 14  acousticness      114000 non-null  float64
 15  instrumentalness  114000 non-null  float64
 16  liveness          114000 non-null  float64
 17  valence           114000 non-null  float64
 18  tempo             114000 non-null  float64
 19  time_signature    114000 non-null  int64
 20  track_genre       114000 non-null  object
dtypes: bool(1), float64(9), int64(6), object(5)
memory usage: 17.5+ MB
```

df.describe()

	Unnamed: 0	popularity	duration_ms	danceability	energy	k
count	114000.000000	114000.000000	1.140000e+05	114000.000000	114000.000000	114000.0000
mean	56999.500000	33.238535	2.280292e+05	0.566800	0.641383	5.30914
std	32909.109681	22.305078	1.072977e+05	0.173542	0.251529	3.55998
min	0.000000	0.000000	0.000000e+00	0.000000	0.000000	0.0000
25%	28499.750000	17.000000	1.740660e+05	0.456000	0.472000	2.0000
50%	56999.500000	35.000000	2.129060e+05	0.580000	0.685000	5.0000
75%	85499.250000	50.000000	2.615060e+05	0.695000	0.854000	8.0000
max	113999.000000	100.000000	5.237295e+06	0.985000	1.000000	11.0000

df.shape
df.columns

```
Index(['Unnamed: 0', 'track_id', 'artists', 'album_name', 'track_name',
       'popularity', 'duration_ms', 'explicit', 'danceability', 'energy',
       'key', 'loudness', 'mode', 'speechiness', 'acousticness',
       'instrumentalness', 'liveness', 'valence', 'tempo', 'time_signature',
       'track_genre'],
      dtype='object')
```

df.isnull().sum()

	0
Unnamed: 0	0
track_id	0
artists	1
album_name	1
track_name	1
popularity	0
duration_ms	0
explicit	0
danceability	0
energy	0
key	0
loudness	0
mode	0
speechiness	0
acousticness	0
instrumentalness	0
liveness	0
valence	0
tempo	0
time_signature	0
track_genre	0

dtype: int64

```
df = df.dropna()
```

```
df = df.drop(['track_id','track_name','artist_name'], axis=1, errors='ignore')
```

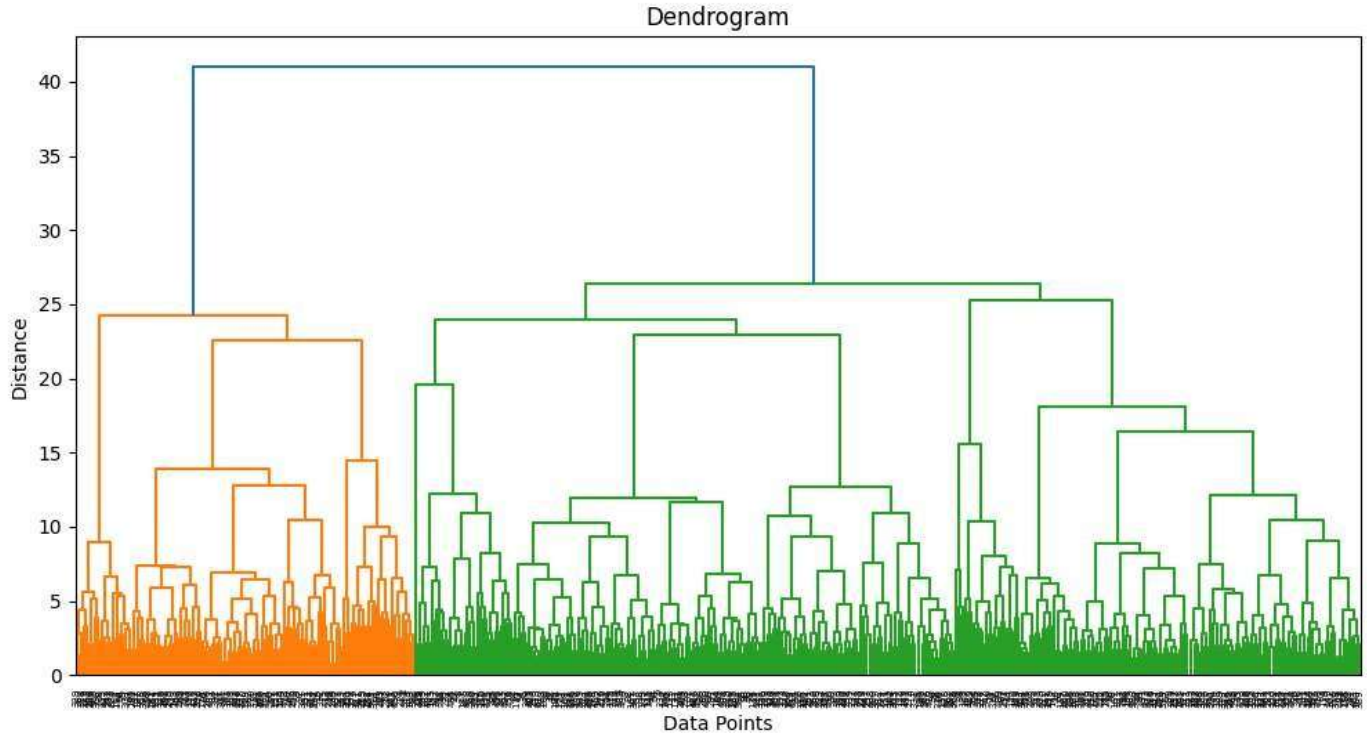
```
df = df.select_dtypes(include=['int64','float64'])
```

```
df = df.sample(n=500, random_state=42)
```

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
```

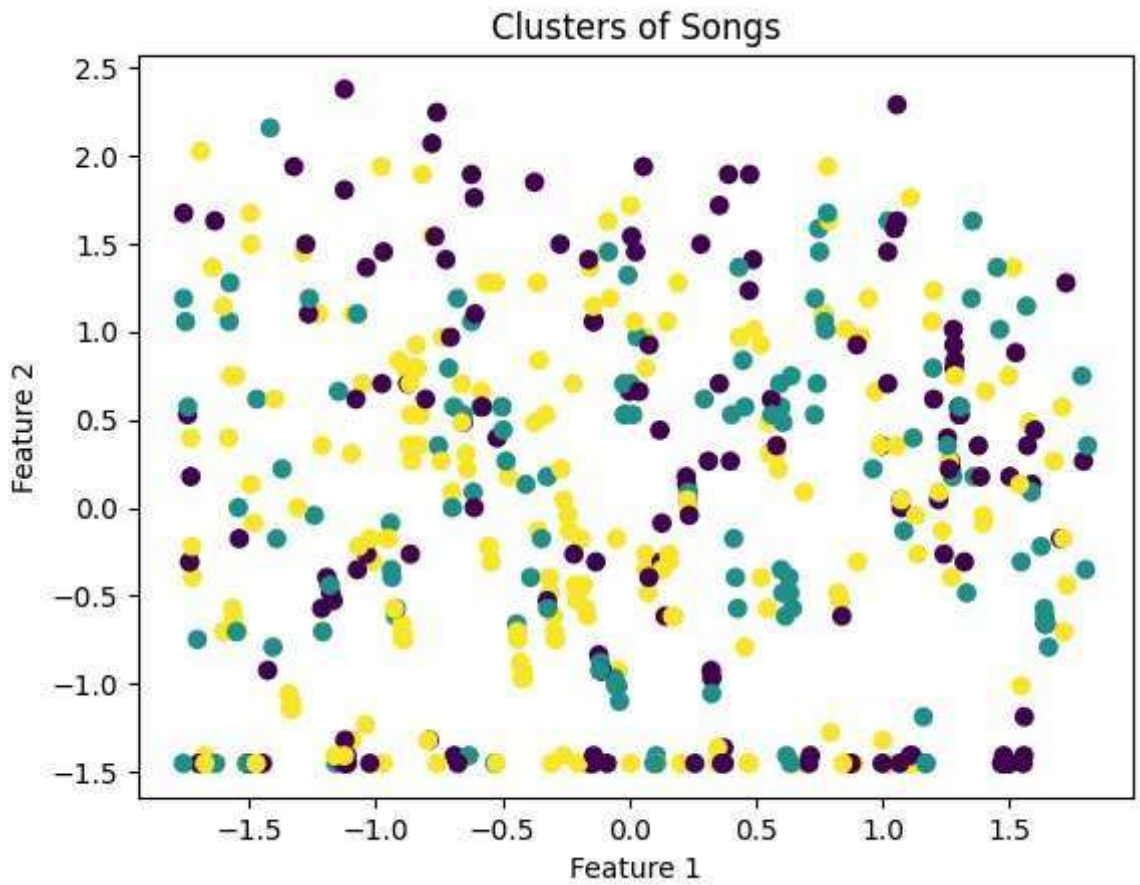
```
scaled_data = scaler.fit_transform(df)
```

```
import scipy.cluster.hierarchy as sch
plt.figure(figsize=(12,6))
sch.dendrogram(sch.linkage(scaled_data, method='ward'))
plt.title("Dendrogram")
plt.xlabel("Data Points")
plt.ylabel("Distance")
plt.show()
```




```
from sklearn.cluster import AgglomerativeClustering
hc = AgglomerativeClustering(n_clusters=3, linkage='ward')
y_pred = hc.fit_predict(scaled_data)
print(y_pred)
```

```
[1 2 1 0 2 2 0 2 2 2 1 2 1 0 1 0 0 2 2 0 0 0 2 0 2 0 0 2 1 2 1 0 2 1 2 2 2
 2 0 1 1 2 2 1 0 0 0 2 2 2 2 2 0 2 2 2 2 0 2 2 1 0 1 1 2 2 0 2 1 0 1 2 2
 0 1 2 2 2 1 2 1 0 0 0 2 2 1 2 2 1 1 1 0 2 1 0 0 2 2 0 0 2 0 1 1 2 0 1 2 2
 1 2 1 0 2 2 2 1 0 1 0 2 2 1 2 0 0 0 2 2 2 2 2 0 2 2 2 0 0 0 1 1 2 2 2 2 0
 0 0 0 0 1 2 1 2 0 2 1 2 1 2 1 0 2 2 2 0 2 0 1 2 2 2 1 2 0 2 2 2 2 0 1 0 1 0
 0 0 1 1 0 2 1 0 1 0 0 2 0 1 2 2 0 2 1 2 0 0 0 1 2 0 0 0 1 1 0 2 0 1 1 2 0
```

```
2 2 0 1 1 2 2 0 2 0 1 1 2 2 1 2 0 2 2 0 0 0 2 1 2 2 0 2 1 2 1 1 1 1 2 2 2
0 0 2 1 0 2 0 1 2 2 1 1 1 2 2 1 0 2 0 0 2 1 1 2 0 1 0 0 2 0 0 2 1 2 1 0 2
0 0 2 0 0 1 1 1 2 1 0 2 0 1 1 0 1 0 0 2 2 0 0 2 2 2 0 1 0 2 1 0 0 1 2 2 2
0 1 2 2 1 2 0 0 2 1 2 2 0 2 1 0 2 2 1 0 2 1 0 2 2 1 1 1 1 2 2 0 2 1 1 2 0
2 1 2 0 2 1 1 0 0 0 0 1 1 2 2 2 0 2 2 0 1 2 1 1 1 2 0 0 2 2 0 2 2 0 2 2 2
2 0 1 2 2 1 2 0 1 0 0 2 0 1 2 0 2 1 2 0 2 2 2 1 2 1 0 2 2 2 2 2 0 1 2 1 0
0 2 0 0 2 1 1 2 2 0 1 1 2 0 2 2 0 0 2 0 0 1 0 1 1 1 2 2 0 0 0 0 1 2 1 0 2
1 2 0 0 2 2 1 0 2 2 0 0 1 2 1 2 1 1 2]
```

```
plt.scatter(scaled_data[:,0], scaled_data[:,1], c=y_pred)
plt.title("Clusters of Songs")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```